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Selecting Optimal Alternatives and Risk Reduction Strategies in Decision Trees

Hanif D. Sherali, Evrim Dalkiran
Grado Department of Industrial and Systems Engineering, Virginia Tech,
Blacksburg, Virginia 24061 {hanifs@vt.edu, dalkiran@vt.edu}

Theodore S. Glickman
Department of Decision Sciences, The George Washington University, Washington, DC 20052, glickman@gwu.edu

In this paper we conduct a quantitative analysis for a strategic risk management problem that involves allocating certain available failure-mitigating and consequence-alleviating resources to reduce the failure probabilities of system safety components and subsequent losses, respectively, together with selecting optimal strategic decision alternatives, to minimize the risk or expected loss in the event of a hazardous occurrence. Using a novel decision tree optimization approach to represent the cascading sequences of probabilistic events as controlled by key decisions and investment alternatives, the problem is modeled as a nonconvex mixed-integer 0-1 factorable program. We develop a specialized branch-and-bound algorithm in which lower bounds are computed via tight linear relaxations of the original problem that are constructed by utilizing a polyhedral outer-approximation mechanism in concert with two alternative linearization schemes having different levels of tightness and complexity. We also suggest three alternative branching schemes, each of which is proven to guarantee convergence to a global optimum for the underlying problem. Extensive computational results and sensitivity analyses are presented to provide insights and to demonstrate the efficacy of the proposed algorithm.

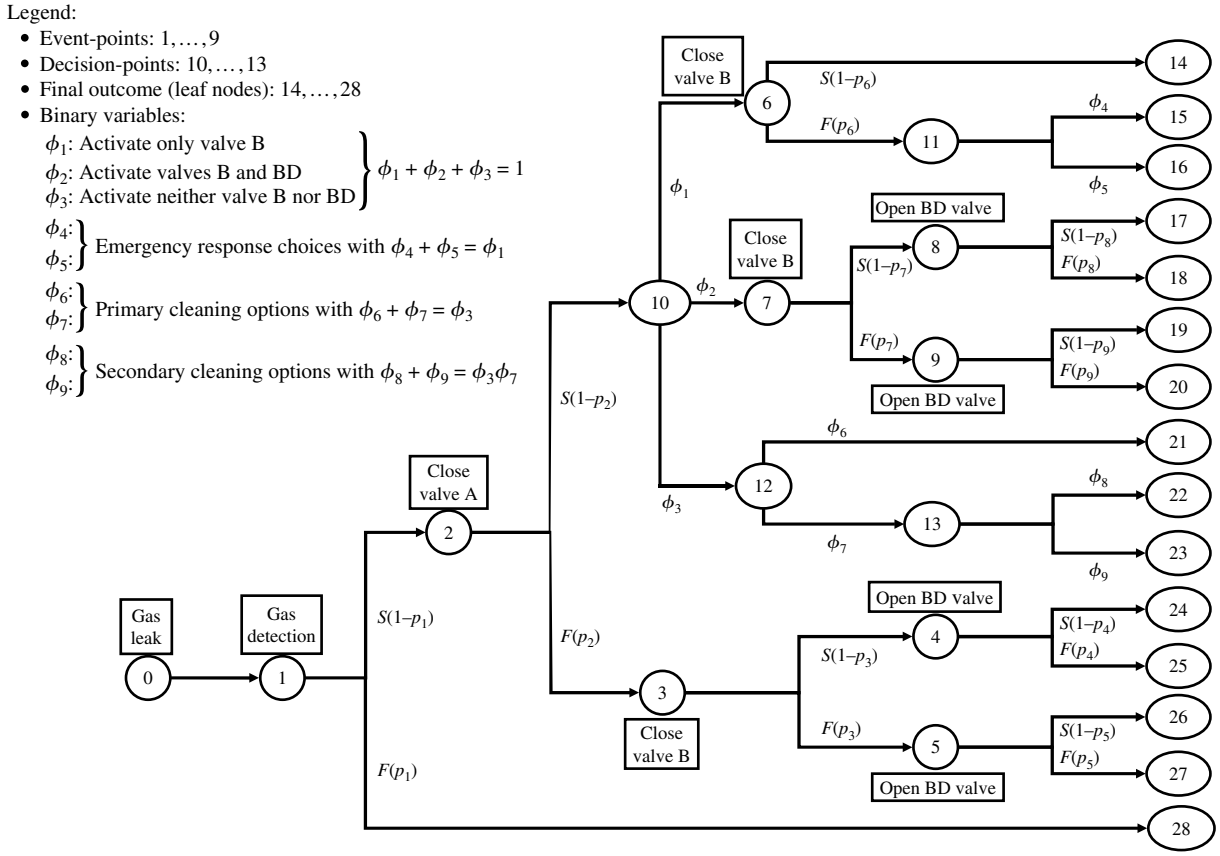
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1. Introduction

This paper addresses the strategic reduction of risk in a system that is characterized by a *decision tree*. Such a tree contains two types of nodes that represent either *event points* or *decision points*. At an event point, some safety feature or measure is invoked that might lead to one of several outcomes, each represented by an arc having a specified probability of occurrence. For example, in the particular context of *Bernoulli events*, an event-point i would trigger either a failure or a success outcome state with respective probabilities p_i and $1 - p_i$ corresponding to the particular safety feature applied at this stage. On the other hand, the arcs emanating from each decision-point node are deterministic and represent a selection among different available alternatives. These choices might involve making certain strategic decisions or system design selections at the particular stage in the process, which then govern the subsequent sequence of events and decisions. The decision tree is rooted at a *root node* or *node zero* that represents some component failure in the system under study or an external hazardous occurrence, which triggers a cascading sequence of strategic decisions and event outcomes based on applied actions or safety features that are invoked to control the damage. Hence, each branch in the tree represents a specific sequence of decision choices and event outcomes starting from the root node and culminating in a *final outcome* or *leaf node* of the tree, where the latter entails an associated *consequence* or *loss*.

Figure 1 displays a particular example of a decision tree pertaining to the rupture of a gas line in an offshore oil and gas platform, which has been adapted from a simpler *event tree* representation (i.e., one having only event-point nodes) as described in Andrews and Dunnett (2000), and that involves Bernoulli events. Here, each event-point, represented by the nodes indexed $1, \ldots, 9$, corresponds to applying some safety measure such as closing an isolation valve to localize the damage or opening a blow-down (BD) valve to depressurize certain critical sections of the system, and can lead to one of two immediate scenarios or outcomes depending on the success or failure of this measure. Note that a particular safety measure can be activated at different event-points in the tree to counteract the hazard at that particular stage in the process. At any decision-point in the tree (nodes $10, \ldots, 13$ in Figure 1), depending on the situation at that particular stage as governed by the sequence of events and decisions leading up to it, one of several strategic alternative options can be selected, where each such option is represented by a binary variable that takes on a value of one if this option is selected and zero otherwise. Hence, at the decision node 10, for example, one of three

Figure 1. A decision tree representing a gas-line rupture.



choices can be made with respect to subsequently activating only the closure of valve B, or the closure of valve B and the opening of the BD valve, or neither of these, which are, respectively, represented by the binary variables ϕ_1, ϕ_2, and ϕ_3, with $\phi_1 + \phi_2 + \phi_3 = 1$. The binary variables designating the alternative choices at the other decision nodes 11, 12, and 13 are specified in the legend of Figure 1, along with their constraining relationships. For example, assume that the first option is chosen at decision node 10. At the event-point represented by node 6, the valve B either fails or succeeds to operate with respective probabilities p_6 and $1 - p_6$. Following the success outcome, the leaf node 14 is reached, whereas the failure event leads to another decision node at which one of two emergency response choices can be made, which are represented by the binary variables ϕ_4 and ϕ_5, respectively, with $\phi_4 + \phi_5 = \phi_1$. Note that when $\phi_1 = 0$, the choice represented at decision node 11 does not arise. In this fashion, the cascading sequences of events and decisions lead to final leaf nodes 14, … , 28, each entailing a particular loss or consequence.

Jiang et al. (2006) study maintenance selection and scheduling under tight budgetary and labor constraints to maximize the risk reduction. A composite heuristic using linear programming relaxations and dynamic programming is proposed to solve the large-scale integer programming problem. In another study, Dillon et al. (2003) develop a decision tool for allocating a limited budget amongst design and residual investments that involve reinforcement (for mitigating technical failures) and initial budget reserves (for mitigating managerial failures). Mehr and Tumer (2006) study resource allocation in the form of a portfolio selection problem in which risk is related to both the likelihood and consequence of an undesirable event, and each unit of resource allocation reduces the risk by a random amount. Sherali et al. (2008) also study resource allocations to minimize the expected cost in a given system. In contrast to Mehr and Tumer (2006), they differentiate between investment decisions for failure mitigation and consequence alleviation. A logit model is used to represent the relationships between investments and failure probabilities instead of adopting a linearity assumption as in Mehr and Tumer (2006). Furthermore, the probability of a final consequence is given via an event tree as a polynomial function of the probabilities of cascading events as opposed to using an assigned probability. The overall event tree optimization problem is modeled as a continuous nonconvex factorable program, and an equivalent transformed reformulation of the problem is solved using the commercial global optimization software BARON (Tawarmalani and Sahinidis 2005).

Several other risk management applications where decision trees of the type described in Figure 1 arise are discussed in the literature. Beim and Hobbs (1997) utilize event trees to estimate the lock closure risks due to vessel accidents and nonstructural failures, whereas Acosta and Siu (1993) analyze steam generator tube-rupture accidents in a power plant. Alternatively, Ulerich and Powers (1988), Andrews and Bartlett (2003), and Hayes (2002) adopt fault trees for modeling chemical processes, firewater deluge systems, and biological invasions, respectively. Dugan et al. (2000) develop a fault tree modeling and analysis tool that decomposes the tree into static and dynamic subtrees that are solved using binary decision diagrams and Markov models, respectively. Another such computer-automated fault tree analysis tool that is applied in chemical process industries has been designed by Khan and Abbasi (2000). For fault tree management problems, Rauzy (1993) proposes new algorithms based on binary decision diagrams that enable an efficient computation of minimal cuts along with deducing the probability of the root event. Dutuit and Rauzy (1996) describe a linear-time algorithm to detect independent subtrees in a fault tree by which computational costs can be reduced by means of a divide-and-conquer technique. Using the binary decision diagram approach of Rauzy (1993) and Dutuit and Rauzy (1996), Sinnamon and Andrews (1997a, b) calculate the exact values of the top event parameters efficiently in contrast with the approximations produced by using traditional kinetic tree theory approaches. In addition, Furuta and Shiraishi (1984), Kenarangui (1991), and Huang et al. (2001) suggest using the fuzzy-set approach to enable the use of verbal statements for possibility measures instead of requiring the specification of event probabilities.

The concept of risk is used in different forms within diverse fields. Similar to the aforementioned papers, we adopt the classical definition of *risk* as the *expected loss* (see Nikulin 2002, Ross 2006) or the expected value of the damage (Kaplan and Garrick 1981), whereas a combination of expected loss and its variance is another common representation of risk, especially in finance. Markowitz (1952, 1987) points out that considering expected returns alone in portfolio optimization fails to sufficiently accommodate diversification, and accordingly proposes the use of mean-variance models in this context. Artzner et al. (1999) indicate that risk is often interpreted in finance as a change in value between two time periods. They define risk in terms of the future net worth of a set of financial instruments at a specific point in time, and they introduce coherent measures of risk that satisfy certain translation invariance, subadditivity, positive homogeneity, and monotonicity properties. Föllmer and Schied (2002) propose alternative convex measures of risk that satisfy the weaker property of convexity in lieu of the foregoing subadditivity and positive homogeneity properties. Ruszczyński and Shapiro (2006) study the optimization of convex risk functions and investigate certain continuity, differentiability, and duality relationships. Note that the expected loss risk measure is coherent, and hence, convex.

The main contributions of this paper are as follows. First, we introduce a novel *decision tree optimization* (DTO) model that significantly extends the event tree model analyzed in Sherali et al. (2008) by accommodating decision nodes having associated multiple options to select from, which enables a strategic restructuring of the tree-based system by way of judiciously composing the operational components of the system. Concurrently, the DTO model considers the allocation of certain available resources within the resulting tree to mitigate failure probabilities and consequence losses to minimize the overall risk. Second, we perform a polyhedral analysis for this more complex model to enhance its solvability by (a) analyzing alternative methods for effectively linearizing the monomials of binary variables within the formulation; (b) exploring suitable polyhedral outer approximations for the region defined by the logarithmic functions in the model; (c) deriving classes of optimality-induced valid inequalities, and (d) designing range reduction strategies for tightening the model representation. Third, we develop a specialized branch-and-bound algorithm that adopts effective alternative partitioning strategies, and we establish its convergence to a global optimum based on several derived theoretical properties of the proposed formulation and its relaxation. Fourth, we present extensive computational results (including insightful sensitivity analyses) using the proposed branch-and-bound algorithm, along with comparisons against the commercial global optimization software BARON, Version 8.1.5 (Tawarmalani and Sahinidis 2005). In particular, we demonstrate a significant improvement in algorithmic performance using the proposed approach.

The remainder of this paper is organized as follows. In §2, we formulate a mathematical model for the DTO problem and then develop a suitable reformulation along with a tight lower-bounding representation through some transformations, polyhedral outer approximations, and valid inequalities in concert with two alternative linearization schemes. In §3, we derive a specialized branch-and-bound procedure to solve the DTO problem and establish its convergence. Sensitivity analyses and computational results are presented in §4 for a hypothetical case study, along with algorithmic performance results and comparisons using a set of simulated problems. Finally, §5 concludes the paper with a summary and recommendations for future extensions.

2. Model Formulation: Analysis and Enhancements

To model the proposed decision tree optimization problem, we begin by defining the risk associated with each possible final outcome or leaf node of the tree. Toward this end, consider the following notation:

- $I_e \equiv \{1, \ldots, I\}$: set of indices representing the event-point nodes.

• p_i and $(1-p_i)$: respectively, the conditional probabilities of failure and success associated with the outcomes resulting from the application of the particular safety feature at event-point i, given that the sequence of decisions and events on the chain from the root node to node i has occurred, $i \in I_e$.

Remark 1. Note that for the sake of notational simplicity and clarity in exposition, we consider Bernoulli events in our model, although our analysis readily generalizes to multistate events. For example, we could consider multiple levels of failure indexed by $r \in F_i$, having respective probabilities p_{ir}, $r \in F_i$, $\forall\, i \in I_e$, with the success probability given by $p_i^s \equiv (1 - \sum_{r \in F_i} p_{ir}) \in (0,1)$. Some comments on accommodating this feature in the model are provided in our discussion.

• $\tau = \{j$: node j is a final outcome or leaf node index of the tree$\}$.

• $d = 1, \ldots, D$: indices for the collective set of decisions made over the entire tree.

• $\phi_d = \begin{cases} 1 & \text{if decision } d \text{ is selected} \\ 0 & \text{otherwise.} \end{cases}$

• K: index set of decision-point nodes in the tree.

• $J_k = \{$decisions d (with corresponding binary variables ϕ_d) associated with the alternative option-based arcs that emanate from node $k\}$, $\forall\, k \in K$.

• $B_k = \{$ordered set of decisions d (with corresponding binary variables ϕ_d) that occur on the (unique) path from the root node 0 to node $k\}$, $\forall\, k \in K \cup I_e \cup \tau$.

For each $j \in \tau$:

• $C_j = \{i$: node i lies on the (unique) path from the root node 0 to node j, excluding nodes 0 and $j\}$.

• $A_{ij} =$ arc emanating from node $i \in C_j$ that lies on the path joining the root node 0 to node $j \in \tau$.

• $S_{1j} = \{i \in C_j \cap I_e$: the (conditional) probability associated with arc A_{ij} is $p_i\}$, $\forall\, j \in \tau$.

• $S_{2j} = \{i \in C_j \cap I_e$: the (conditional) probability associated with arc A_{ij} is $(1-p_i)\}$, $\forall\, j \in \tau$.

• $S_{3j} = \{d$: decision d is associated with arc A_{ij}, $\forall\, i \in C_j \cap K\}$, $\forall\, j \in \tau$.

• $l_j =$ loss or consequence (in dollars, say) associated with leaf node $j \in \tau$.

• $c_d =$ fixed cost associated with selecting decision d, $\forall\, d = 1, \ldots, D$. (Note that for the sake of simplicity, we consider here only linear cost terms; however, the methodology described below readily extends to the case where we might have polynomial cost terms that are predicated on products of the ϕ-variables that occur along (partial) paths from node 0 to any leaf node.)

Then the *risk* associated with the leaf node $j \in \tau$ is the *expected consequence/loss* that is conditioned on the binary decisions as given by

$$\psi_j \equiv l_j \prod_{i \in S_{1j}} p_i \prod_{i \in S_{2j}} (1-p_i) \prod_{d \in S_{3j}} \phi_d, \quad \forall\, j \in \tau, \tag{1}$$

and the *overall risk* is the total expected consequence given by $\sum_{j \in \tau} \psi_j$.

Thus far, the only decisions identified with respect to the decision tree analysis involve selecting binary values for the variables ϕ_d, $d = 1, \ldots, D$, subject to the restrictions that

$$\sum_{d \in J_k} \phi_d = \prod_{d \in B_k} \phi_d, \quad \forall\, k \in K, \tag{2}$$

where $\prod_{d \in B_k} \phi_d \equiv 1$ whenever $B_k = \varnothing$, $\forall\, k \in K$. Now, as in Sherali et al. (2008), suppose that we can additionally bring to bear certain preventive or protective *event-related resources* such as investments in improved technologies or supporting equipment for the various safety features involved so as to mitigate the conditional failure probabilities, p_i, $i \in I_e$, given the sequence of preceding actions and events. Likewise, suppose that we can invest in certain available *consequence-related resources* such as clean-up mechanisms and trained emergency response personnel to ameliorate the potential consequences or losses l_j associated with the leaf nodes $j \in \tau$. Accordingly, the goal would be to determine how to effectively deploy the available limited resources under some budgetary restrictions so as to appropriately manipulate the failure probabilities and the resultant consequences, as well as devise a plan for making suitable strategic decision option choices, to minimize the overall risk. More specifically, consider the following related notation and concepts:

• $m = 1, \ldots, M$: index set for the available event-related resources.

• $s_m =$ total available units of the event-related resource m, $\forall\, m = 1, \ldots, M$.

• $q_{im} =$ *decision variable* representing the quantity of event-related resource m that is allocated to reduce the associated (conditional) failure probability p_i of the safety feature deployed at event-point or node i, $\forall\, i \in I_e$, $m = 1, \ldots, M$.

• $c_{im} =$ cost (in dollars) per unit of q_{im}.

• $n = 1, \ldots, N$: index set for the available-consequence-related resources.

• $t_n =$ total available units of the consequence-related resource n, $\forall\, n = 1, \ldots, N$.

• $r_{jn} =$ *decision variable* representing the quantity of consequence-related resource n that is allocated to reduce the loss magnitude l_j associated with the leaf node j, $\forall\, j \in \tau$, $n = 1, \ldots, N$.

• $d_{jn} =$ cost (in dollars) per unit of r_{jn}.

• $\beta =$ total available budget (in dollars).

• *Logit model for relating* p_i *to* $\{q_{im}, m = 1, \ldots, M\}$, $\forall\, i \in I_e$*:*

$$\ln\left[\frac{p_i}{1-p_i}\right] = a_{i0} - \sum_{m=1}^{M} a_{im} q_{im}, \quad \forall\, i \in I_e, \text{ where } (a_{i0}, \ldots, a_{iM}) \geqslant 0. \tag{3}$$

• $[p_i^l, p_i^u]$: lower and upper bounds on p_i, where $0 < p_i^l \leqslant p_i \leqslant p_i^u < 1, \ \forall i \in I_e$, and where these bounds are either imposed or are implied by (3) and the available resources; in particular, we assume that $p_i^u = p_i^{u(3)} \equiv e^{a_{i0}}/(1+e^{a_{i0}}), \forall i \in I_e$, as implied by (3), i.e., the model seeks further possible reductions in p_i below p_i^u by using suitable event-related resource allocations as necessary.

• *Model for relating l_j to $\{r_{jn}, n = 1, \ldots, N\}, \ \forall j \in \tau$:*

$$l_j = b_{j0} - \sum_{n=1}^{N} b_{jn} r_{jn}, \quad \forall j \in \tau, \text{ where } (b_{j0}, \ldots, b_{jN}) \geqslant 0. \quad (4)$$

Remark 2. As an alternative to (4), we could consider a diminishing marginal return loss function of the type $l_j = b_{j0} \exp\{-\sum_{n=1}^{N} b_{jn} r_{jn}\}, \ \forall j \in \tau$, having nonnegative parameter values. Our proposed algorithm can be identically applied for such loss functions upon taking logarithms. Likewise, for the general multifailure state scenario discussed in Remark 1, the corresponding logit model would take the form $\ln[p_{ir}/p_i^s] = a_{ir0} - \sum_{m=1}^{M} a_{irm} q_{im}, \ \forall r \in F_i, i \in I_e$, with the analysis described below following identically.

• $[l_j^l, l_j^u]$: lower and upper bounds on l_j, where $0 < l_j^l \leqslant l_j \leqslant l_j^u < \infty, \ \forall j \in \tau$, and where these bounds are either imposed or are implied by (4) and the available resources; in particular, we assume that $l_j^u = l_j^{u(4)} \equiv b_{j0}, \ \forall j \in \tau$, as implied by (4), i.e., the model seeks further possible reductions in l_j below l_j^u by using suitable consequence-related resource allocations as necessary.

The *decision tree optimization* problem, **DTO**, can then be formulated as the following 0-1 mixed-integer nonlinear programming problem:

DTO:

$$\text{minimize} \quad \sum_{d=1}^{D} c_d \phi_d + \sum_{j \in \tau} l_j \prod_{i \in S_{1j}} p_i \prod_{i \in S_{2j}} (1 - p_i) \prod_{d \in S_{3j}} \phi_d \quad (5a)$$

$$\text{subject to} \quad \sum_{i \in I_e} q_{im} \leqslant s_m, \quad \forall m = 1, \ldots, M \quad (5b)$$

$$\sum_{j \in \tau} r_{jn} \leqslant t_n, \quad \forall n = 1, \ldots, N \quad (5c)$$

$$\sum_{i \in I_e} \sum_{m=1}^{M} c_{im} q_{im} + \sum_{j \in \tau} \sum_{n=1}^{N} d_{jn} r_{jn} \leqslant \beta, \quad (5d)$$

$$\ln\left[\frac{p_i}{1-p_i}\right] = a_{i0} - \sum_{m=1}^{M} a_{im} q_{im}, \quad \forall i \in I_e \quad (5e)$$

$$l_j = b_{0j} - \sum_{n=1}^{N} b_{jn} r_{jn}, \quad \forall j \in \tau \quad (5f)$$

$$\sum_{d \in J_k} \phi_d = \prod_{d \in B_k} \phi_d, \quad \forall k \in K \quad (5g)$$

$$(p, l) \in \Omega \equiv \left\{ (p, l): \begin{array}{c} 0 < p_i^l \leqslant p_i \leqslant p_i^u < 1, \\ \forall i \in I_e \\ 0 < l_j^l \leqslant l_j \leqslant l_j^u < \infty, \\ \forall j \in \tau \end{array} \right\}, \quad (5h)$$

$$(q, r) \in \mathcal{P} \quad (5i)$$

$$\phi_d \text{ binary}, \quad \forall d = 1, \ldots, D. \quad (5j)$$

In this formulation, the objective function (5a) seeks to minimize the total cost of implementing decisions plus the consequent expected loss or overall risk. (Note that the first term in the objective function is equivalent to the polynomial term $\sum_{k \in K} \sum_{d \in J_k} c_d \phi_d [\prod_{d' \in B_k} \phi_{d'}]$ because of constraints (5g) and (5j).) Whereas we adopt the objective function (5a) in this paper, we could alternatively consider a mean-variance model following Markowitz (1952, 1987) by accommodating a variance term within (5a), where the variance of loss is given by

$$\text{Var(loss)} = \sum_{j \in \tau} l_j^2 \prod_{i \in S_{1j}} p_i \prod_{i \in S_{2j}} (1 - p_i) \prod_{d \in S_{3j}} \phi_d - \left(\sum_{j \in \tau} l_j \prod_{i \in S_{1j}} p_i \prod_{i \in S_{2j}} (1 - p_i) \prod_{d \in S_{3j}} \phi_d \right)^2. \quad (6)$$

Although this complicates the model by doubling the degree of the polynomial objective function, the reformulation mechanism and polyhedral analysis presented below can readily handle the variance term (6) by introducing $|\tau| + 2$ new variables along with suitable linearization constructs. However, we shall focus on (5a) in this study, and we propose using the mean-variance risk function as a future extension. Constraints (5b) and (5c) impose the resource availability restrictions; Constraint (5d) enforces the budgetary limitation; constraints (5e), (5f), and (5g) follow from (3), (4), and (2), respectively; (5h) requires the (p, l)-variables to satisfy the specified bounding restrictions; and $\mathcal{P}$ in (5i) is a polyhedron (embedded in the nonnegative orthant) that includes any additional suitable constraints and variables in order to further restrict the (q, r)-variables, or to relate them to other influencing technological or operational decisions that are not explicitly stated in the above model. In the simplest case, $\mathcal{P} \equiv \{(q, r): (q, r) \geqslant 0\}$. Observe that the nonconvexity in Problem DTO arises due to the polynomial function in (5a) and (5g), the logarithmic (factorable) term in (5e), and as binary decision variables in (5j).

Remark 3. There is another related model that can be formulated to study a sensitivity analysis issue in the context of decision trees, which is also of interest and can be handled by the algorithmic process discussed below. Consider a decision tree in which the event probabilities and loss values are not known with certainty but might vary within intervals as designated in (5h) (while not being strategically controllable). Given such variabilities in the probabilities and the consequences, we might be interested in ascertaining the least expected consequence or risk value, along with the maximum possible deviation from this value that could occur due to the inherent uncertainties in the problem. Hence, we could first minimize (5a) subject to

(5g), (5h), and (5j), and then, fixing the resulting ϕ-variable values, we could next determine the maximum of (5a) subject to (5h) to resolve these two respective issues. Furthermore, as an alternative approach for making decisions that hedge against this uncertainty, we could formulate and solve a corresponding robust optimization problem in the spirit of Mulvey et al. (1995) or minimize the maximum risk by solving $\min_{\phi:(5g),(5j)} \max_{(p,l)\in\Omega}\{(5a)\}$.

We next define the following auxiliary variables, along with their implied bounds, to conveniently reformulate Problem DTO. To begin with, we transform the objective function by denoting

$$\theta_j \equiv l_j \prod_{i\in S_{1j}} p_i \prod_{i\in S_{2j}} (1-p_i), \quad \forall\, j \in \tau. \tag{7a}$$

Thus, we have

$$\theta_j^l \leqslant \theta_j \leqslant \theta_j^u, \quad \text{where} \left\{ \begin{array}{l} \theta_j^l = l_j^l \prod_{i\in S_{1j}} p_i^l \prod_{i\in S_{2j}} (1-p_i^u) \\ \theta_j^u = l_j^u \prod_{i\in S_{1j}} p_i^u \prod_{i\in S_{2j}} (1-p_i^l) \end{array} \right\}, \quad \forall\, j \in \tau. \tag{7b}$$

Furthermore, noting (1), (5a), and (7a), we denote

$$\psi_j \equiv \theta_j w_{S_{3j}}, \quad \forall\, j \in \tau, \tag{7c}$$

where

$$w_{S_{3j}} \equiv \prod_{d\in S_{3j}} \phi_d, \quad \forall\, j \in \tau. \tag{7d}$$

Noting that the binariness of the ϕ_d-variables implies that of the $w_{S_{3j}}$-variables, $\forall\, j \in \tau$, we can linearize (7c) by replacing it with the restrictions

$$\begin{aligned} &\theta_j^l w_{S_{3j}} \leqslant \psi_j \leqslant \theta_j^u w_{S_{3j}} \quad \text{and} \\ &\theta_j^l (1 - w_{S_{3j}}) \leqslant \theta_j - \psi_j \leqslant \theta_j^u (1 - w_{S_{3j}}), \end{aligned} \tag{7e}$$

by which it is readily verified that (7c) holds true whenever $w_{S_{3j}}$ takes on binary values, $\forall\, j \in \tau$. In fact, because of the positive (unit) objective coefficients on ψ_j, this relationship will hold true even if we retain just the inequalities $\psi_j \geqslant \theta_j^l w_{S_{3j}}$ and $\psi_j \geqslant \theta_j - \theta_j^u (1 - w_{S_{3j}})$ from (7e), $\forall\, j \in \tau$.

In the same spirit as $w_{S_{3j}}, j \in \tau$, we define a new variable w_J to represent $\prod_{d\in J} \phi_d$ for some specific sets $J \subset \{1, \ldots, D\}$ as described below, which will be used in the sequel for linearizing (5g). Here, whenever $|J| = 1$ with $J = \{d\}$, we will take the corresponding $w_J \equiv \phi_d$ itself, and whenever $J = \varnothing$, we will take $w_J \equiv 1$. Now consider any set S_{3j} for $j \in \tau$ (in general, $w_{S_{3j}}$ might be replicated for different $j \in \tau$), and assume that the indices in S_{3j} are arranged according to the order in which the corresponding decisions d (with associated variables ϕ_d) occur along the path from the root node to node j. If $|S_{3j}| \geqslant 2$, this will generate w_J-variables via sets J defined by taking the first two indices from S_{3j}, the first three indices from S_{3j}, and so on, up to all the indices from S_{3j} (finally yielding $w_{S_{3j}}$ as denoted above). Letting $\mathcal{J}$ denote the resulting distinct sets J generated, we define the following identities:

$$w_J = \prod_{d\in J} \phi_d, \quad \forall\, J \in \mathcal{J}. \tag{8}$$

Note that (8) subsumes (7d) because $S_{3j} \in \mathcal{J},\ \forall\, j \in \tau$.

Next, to linearize (5e), we introduce the variables

$$y_{1i} = \ln(p_i) \quad \text{and} \quad y_{2i} = \ln(1 - p_i),\ \forall\, i \in I_e. \tag{9a}$$

Note that

$$y_{1i}^l \leqslant y_{1i} \leqslant y_{1i}^u \quad \text{and} \quad y_{2i}^l \leqslant y_{2i} \leqslant y_{2i}^u,$$

where

$$\begin{aligned} &y_{1i}^l = \ln(p_i^l), \quad y_{1i}^u = \ln(p_i^u), \quad y_{2i}^l = \ln(1 - p_i^u), \quad \text{and} \\ &y_{2i}^u = \ln(1 - p_i^l), \quad \forall\, i \in I_e. \end{aligned} \tag{9b}$$

Similarly, to linearize (7a) itself, we denote

$$z_j \equiv \ln(\theta_j), \quad \forall\, j \in \tau, \tag{10a}$$

where, based on (7b), we impose

$$\begin{aligned} &z_j^l \leqslant z_j \leqslant z_j^u, \quad \text{with } z_j^l = \ln(\theta_j^l) \quad \text{and} \\ &z_j^u = \ln(\theta_j^u), \quad \forall\, j \in \tau. \end{aligned} \tag{10b}$$

Likewise, to accommodate the term $\ln(l_j)$ generated by taking logarithms in (7a), define

$$\xi_j \equiv \ln(l_j), \quad \forall\, j \in \tau, \tag{11a}$$

and impose the related bounds

$$\begin{aligned} &\xi_j^l \leqslant \xi_j \leqslant \xi_j^u, \quad \text{where} \quad \xi_j^l = \ln(l_j^l) \quad \text{and} \\ &\xi_j^u = \ln(l_j^u), \quad \forall\, j \in \tau. \end{aligned} \tag{11b}$$

Using the foregoing transformations and substitutions, we obtain the following equivalently reformulated problem DTO(Ω), which is predicated on the hyperrectangle Ω.

$$\textbf{DTO}(\Omega): \quad \text{minimize} \quad \sum_{d=1}^{D} c_d \phi_d + \sum_{j\in\tau} \psi_j \tag{12a}$$

subject to

$$\psi_j \geqslant \theta_j^l w_{S_{3j}} \quad \text{and} \quad \psi_j \geqslant \theta_j - \theta_j^u (1 - w_{S_{3j}}), \quad \forall\, j \in \tau \tag{12b}$$

$$w_J = \prod_{d\in J} \phi_d, \quad \forall\, J \in \mathcal{J} \tag{12c}$$

$$\sum_{i\in I_e} q_{im} \leqslant s_m, \quad \forall\, m = 1, \ldots, M \tag{12d}$$

$$\sum_{j\in\tau} r_{jn} \leqslant t_n, \quad \forall\, n = 1, \ldots, N \tag{12e}$$

$$\sum_{i\in I_e}\sum_{m=1}^{M} c_{im}q_{im} + \sum_{j\in\tau}\sum_{n=1}^{N} d_{jn}r_{jn} \leqslant \beta, \tag{12f}$$

$$y_{1i} - y_{2i} = a_{i0} - \sum_{m=1}^{M} a_{im}q_{im}, \quad \forall\, i \in I_e \tag{12g}$$

$$l_j = b_{0j} - \sum_{n=1}^{N} b_{jn}r_{jn}, \quad \forall\, j \in \tau \tag{12h}$$

$$\sum_{d\in J_k} \phi_d = w_{B_k}, \quad \forall\, k \in K \tag{12i}$$

$$z_j = \xi_j + \sum_{i\in S_{1j}} y_{1i} + \sum_{i\in S_{2j}} y_{2i}, \quad \forall\, j \in \tau \tag{12j}$$

$$y_{1i} = \ln(p_i), \quad \forall\, i \in I_e \tag{12k}$$

$$y_{2i} = \ln(1 - p_i), \quad \forall\, i \in I_e \tag{12l}$$

$$z_j = \ln(\theta_j), \quad \forall\, j \in \tau \tag{12m}$$

$$\xi_j = \ln(l_j), \quad \forall\, j \in \tau \tag{12n}$$

$$(p, l) \in \Omega \tag{12o}$$

$$(q, r) \in \mathcal{P} \tag{12p}$$

$$\left.\begin{array}{l} \theta_j^l \leqslant \theta_j \leqslant \theta_j^u,\ z_j^l \leqslant z_j \leqslant z_j^u, \text{ and } \xi_j^l \leqslant \xi_j \leqslant \xi_j^u, \\ \qquad\qquad\qquad\qquad\qquad\qquad \forall\, j \in \tau \\ y_{1i}^l \leqslant y_{1i} \leqslant y_{1i}^u \text{ and } y_{2i}^l \leqslant y_{2i} \leqslant y_{2i}^u, \ \forall\, i \in I_e \end{array}\right\} \tag{12q}$$

$$\begin{array}{l} \phi_d \text{ binary}, \quad \forall\, d = 1, \ldots, D, \quad \text{and} \\ 0 \leqslant w_J \leqslant 1, \quad \forall\, J \in \mathcal{J}. \end{array} \tag{12r}$$

Here, Ω is as specified in (5h), and we note that the bounds in (12q) depend on Ω (even as Ω will be modified via a partitioning process in our proposed algorithmic approach below) and are given by (7b), (9b), (10b), and (11b), respectively. For the sake of convenience in discussion, we shall denote the set of variables in Problem DTO(Ω), with obvious vector notation, as

$$x \equiv (p, l, q, r, \phi, w, \psi, \theta, y_1, y_2, z, \xi),$$

where, in particular, $y_1 \equiv (y_{1i}, \forall\, i \in I_e)$ and $y_2 \equiv (y_{2i}, \forall\, i \in I_e)$. Observe that DTO($\Omega$) is, in general, a mixed-integer 0-1 factorable program (see Sherali and Wang 2001 for continuous factorable programs). However, in our case, DTO(Ω) is linear except for the complicating identities (12c) and (12k)–(12n).

We next discuss some suitable polyhedral outer-approximation mechanisms to handle the univariate, monotone logarithmic functions in (12k)–(12n). Toward this end, generically denote any identity in (12k)–(12n) as

$$f = \ln(\gamma), \quad \text{where } 0 < \gamma^l \leqslant \gamma \leqslant \gamma^u < \infty. \tag{13}$$

We then replace each such constraint (13) by the following affine convex envelope:

$$f \geqslant \ln(\gamma^l) + \frac{(\gamma - \gamma^l)}{(\gamma^u - \gamma^l)}[\ln(\gamma^u) - \ln(\gamma^l)], \tag{14a}$$

along with some $H \geqslant 2$ tangential supports

$$f \leqslant \ln(\gamma_h) + \frac{(\gamma - \gamma_h)}{\gamma_h}, \quad \text{for } h = 1, \ldots H, \quad \text{where } \gamma^l \equiv \gamma_1 < \gamma_2 < \cdots < \gamma_H \equiv \gamma^u. \tag{14b}$$

To prescribe some judicious alternatives for selecting the points of tangency $\{\gamma_2, \ldots, \gamma_{H-1}\}$ (other than the interval endpoints) in (14b), consider the following result:

Proposition 1. *Consider any $0 < \bar\gamma < \hat\gamma$ such that $\ln(\hat\gamma) - \ln(\bar\gamma) = \Delta$. Then, for $\gamma \in [\bar\gamma, \hat\gamma]$, the maximum error, E, between the function $\ln(\gamma)$ and its piecewise-linear approximation defined by the tangential supports at $\bar\gamma$ and $\hat\gamma$ depends on Δ alone and is given by*

$$E = \ln[e^{\Delta} - 1] - \ln(\Delta) + \frac{\Delta}{e^{\Delta} - 1} - 1. \tag{15}$$

Proof. By the monotonicity of both the logarithmic and the affine tangential supporting functions, the stated maximum error, E, occurs at the point of intersection γ^* of the tangential supports as given by $\gamma^* = \bar\gamma\hat\gamma\Delta/(\hat\gamma - \bar\gamma)$, with

$$E = \frac{\hat\gamma \ln(\hat\gamma) - \bar\gamma \ln(\bar\gamma)}{(\hat\gamma - \bar\gamma)} - 1 - \ln(\gamma^*). \tag{16}$$

Substituting for γ^* in (16), and writing $\ln(\hat\gamma) = \ln(\bar\gamma) + \Delta$, we get

$$E = \ln\left(\frac{\hat\gamma}{\bar\gamma} - 1\right) - \ln(\Delta) + \frac{\Delta}{[(\hat\gamma/\bar\gamma) - 1]} - 1,$$

which, upon using $\hat\gamma/\bar\gamma = e^{\Delta}$, yields (15). □

Corollary 1. *Given any $\epsilon > 0$, let $\Delta = \Delta_\epsilon$ be the solution to* (15) *when we set $E = \epsilon$. Now, suppose that we approximate the function $f \equiv \ln(\gamma)$ on $[\gamma^l, \gamma^u] \subseteq (0, \infty)$ by H tangential supports constructed at points uniformly distributed along the f-axis, including the endpoints, where*

$$H = \left\lceil \frac{\ln(\gamma^u) - \ln(\gamma^l)}{\Delta_\epsilon} \right\rceil + 1. \tag{17}$$

Then, the maximum approximation error will be bounded above by ϵ.

Proof. By Proposition 1 and because $\partial E/\partial\Delta > 0$ for $\Delta > 0$ in (15), the maximum approximation error will be bounded above by ϵ provided $[\ln(\gamma^u) - \ln(\gamma^l)]/(H-1) \leqslant \Delta_\epsilon$, which yields (17). □

Accordingly, we define the *Bounded Error Strategy* (**BES**) for selecting points for generating the tangential supports (14b) as that prescribed by Corollary 1, given any error tolerance $\epsilon > 0$. The number of tangential supports generated would then further depend on the bounding interval $[\gamma^l, \gamma^u]$ as given by (17). Alternatively, to control the number of tangential supports generated, we shall also apply BES with a prespecified value of H. In this

case, we generate (14b) at some H uniformly distributed points along the f-axis, including the interval endpoints. This yields

$$\gamma_h = \exp\left\{\ln(\gamma^l) + \left(\frac{h-1}{H-1}\right)\ln\left[\frac{\gamma^u}{\gamma^l}\right]\right\}, \quad \text{for } h = 1, \ldots, H.$$

We shall refer to this strategy as **BES**(H). We recommend the value $H = 4$ based on our computational results reported in §4, where we investigated using $H = 4, \ldots, 20$.

Next, to generate a linear programming relaxation LP(Ω) for computing lower bounds in a branch-and-bound framework, we additionally adopt the following alternative linearization strategies for (12c) as identified in §2.1 below.

2.1. Linearization of (12c)

In this section, we shall discuss two alternative schemes for linearizing (12c) that differ in their size or complexity and the relative tightness of the resulting LP relaxation. These are, respectively, denoted as linearization methods LM1 and LM2, and, together with the polyhedral outer-approximation (14) applied to (12k)–(12n), produce corresponding LP relaxations LP1(Ω) and LP2(Ω), respectively. We shall also use the terminology LP(Ω) to refer generically to either of the foregoing relaxations LP1(Ω) and LP2(Ω).

2.1.1. Linearization Method LM1. This is a standard linearization technique that utilizes the following constraints for each $J \in \mathcal{J}$ such that $|J| \geqslant 2$ (noting (12r)):

$$w_J \leqslant \phi_d, \quad \forall\, d \in J, \quad \text{and} \quad w_J \geqslant \sum_{d \in J} \phi_d - |J| + 1. \tag{18}$$

Note that as portended by (12c), and using (12r), when $\phi_d = 0$ for any $d \in J$, then (18) implies that $w_J = 0$, whereas when $\phi_d = 1$, $\forall\, d \in J$, then (18) implies that $w_J = 1$ as well.

2.1.2. Linearization Method LM2. In this method, consider any decision-point node $k \in K$. Recall that B_k denotes the ordered set of decision indices that occur on the path from node 0 to node k. If $B_k = \varnothing$, then no additional constraints are generated for this node k. Else, suppose that $|B_k| \geqslant 1$. Then, we generate the restrictions

$$w_{B_k + d} = \phi_d, \quad \forall\, d \in J_k \text{ (for each } k \in K\text{: } B_k \neq \varnothing), \tag{19}$$

where we denote $w_{B_k+d} \equiv w_{B_k \cup \{d\}}$, with the index d appearing last in the resulting set $B_k \cup \{d\}$, $\forall\, d \in J_k$.

Proposition 2. *The constraints* (19) *are valid and, together with* (12i) *and* (12r)*, imply that* (12c) *holds true.*

Proof. First of all, note that (19) is valid because if $\phi_d = 0$ for any $d \in J_k$, then we must have $w_{B_k+d} = 0$ by its interpretation, and if $\phi_d = 1$ for any $d \in J_k$, then by (12i) and (12r), we must have $w_{B_k} = 1$, so that again the interpretation of w_{B_k+d} implies that we must have $w_{B_k+d} = 1$.

Next, let us establish that (12c) holds true by induction on $|J|$. Consider any $k \in K$ such that $B_k = \{d'\}$ and $d \in J_k$ so that $|J| = 2$ with $J \equiv \{d', d\}$ (the case $|J| = 1$ is trivial by definition). Now, if $\phi_{d'} = 0$, then (12i) implies that $\phi_d = 0$, so that (19) yields $w_{d'd} = 0$. On the other hand, if $\phi_{d'} = 1$, then (12i) implies that $\phi_d = 0$ or 1, which, respectively, yields $w_{d'd} = 0$ or 1 via (19).

Inductively, to complete the proof, consider $J = \{d_1, \ldots, d_h\}$, where $h \geqslant 3$, and assume that (12c) holds true for any strict subset of this set. Hence, there exists $k \in K$ such that $B_k = \{d_1, \ldots, d_{h-1}\}$, with $d_h \in J_k$. Let us show that (12c) is satisfied for any such J. If $\phi_d = 0$ for any $d \in \{d_1, \ldots, d_{h-1}\}$, then by the induction hypothesis, we have that $w_{B_k} = 0$, which implies by (12i) that $\phi_d = 0$, $\forall\, d \in J_k$. Therefore, $\phi_{d_h} = 0$, which implies by (19) that $w_J = 0$. Else, if $\phi_{d_1} = \cdots = \phi_{d_{h-1}} = 1$, then $w_{B_k} = 1$ by the induction hypothesis, and so by (12i), $\phi_{d_h} = 0$ or 1, which, respectively, yields $w_J = 0$ or 1 via (19), and so (12c) is again satisfied in either case. □

The next result demonstrates that LM2 yields a *tighter representation* than LM1, i.e., its defining constraints imply those of LM1 in the continuous (LP) sense.

Proposition 3. *The constraints represented in* (19)*,* (12i)*, and* $\phi \geqslant 0$ *of LM*2 *imply those in* (18) *of LM*1*.*

Proof. Consider any $J' \in \mathcal{J}$, and assume without loss of generality that $J' = \{1, \ldots, h\}$, with $\phi_1, \ldots, \phi_h$ occurring in this order along the path from node 0 to some node j. Hence, by (18), LM1 generates the constraints

$$w_{J'} \leqslant \phi_d, \quad \forall\, d = 1, \ldots, h, \tag{20a}$$

$$w_{J'} \geqslant \sum_{d=1}^{h} \phi_d - (h-1). \tag{20b}$$

Now, by (19) applied to the node from which the arc having ϕ_h emanates, LM2 directly produces $w_{J'} = \phi_h$ and (12i) yields $\phi_h \leqslant w_{J'-\{h\}}$, which, by (19) applied at the node from which the arc having the variable ϕ_{h-1} emanates, produces $w_{J'-\{h\}} = \phi_{h-1}$. Hence, $w_{J'} \leqslant \phi_{h-1}$ as well. Continuing in this fashion along the chain from node j to node 0 yields that (20a) is implied.

To complete the proof, we next show that (20b) is implied as well. Note that (19) yields for any $j_1 \in J_k$, $k \in K$, that

$$w_{B_k + j_1} = \phi_{j_1} \geqslant w_{B_k} + \phi_{j_1} - 1, \tag{21}$$

where the last inequality follows from the fact that $w_{B_k} \leqslant 1$ in (12r). Hence, applying (21) at the node from which the arc having the variable ϕ_h emanates with $B_k + j_1 \equiv J'$ and $j_1 \equiv h$, we get

$$w_{J'} \geqslant w_{J'-\{h\}} + \phi_h - 1. \tag{22}$$

Repeating this at the node from which the arc having the variable ϕ_{h-1} emanates yields $w_{J'-\{h\}} \geqslant w_{J'-\{h,h-1\}} + \phi_{h-1} - 1$, which, together with (22), implies that

$$w_{J'} \geqslant w_{J'-\{h,h-1\}} + \phi_h + \phi_{h-1} - 2. \tag{23}$$

Continuing (23) along the chain from node j to node 0, we get

$$w_{J'} \geqslant w_{J'-\{h,\,h-1,\ldots,2\}} + \phi_h + \phi_{h-1} + \cdots + \phi_2 - (h-1)$$
$$= \sum_{d=1}^{h} \phi_d - (h-1),$$

where the last equality follows by noting that $w_{J'-\{h,\,h-1,\ldots,2\}} = w_1 \equiv \phi_1$. Hence, (20b) is also implied. □

Now, for any $j \in K \cup I_e \cup \tau$, let $d(j)$ be the first decision index that is encountered in the (reverse) path from node j to the root node (whenever $d(j)$ does not exist, we define $\phi_{d(j)} \equiv 1$). Then we can simplify LM2 by eliminating the w_J-variables upon using (19), which effectively equates $w_{S_{3j}} = \phi_{d(j)},\ \forall\, j \in \tau$ in (12b) and $w_{B_k} = \phi_{d(k)},\ \forall\, k \in K$ in (12i). Hence, by Proposition 2, LM2 *can be equivalently written by using these substitutions in* (12b) *and* (12i)*, and eliminating* (12c) *and the* w_J*-inequalities in* (12r). The following result reveals a partial convex hull representation inherent within LM2.

Proposition 4. *Consider the following polyhedral set defined by the constraints of LM2.*

$$\Lambda \equiv \left\{ \phi \geqslant 0\colon \sum_{d \in J_k} \phi_d = \phi_{d(k)},\ \forall\, k \in K \right\}. \tag{24}$$

Then, Λ *is nonempty and compact with binary-valued extreme points.*

Proof. The compactness of Λ follows readily by noting that the constraints in Λ imply that $0 \leqslant \phi_d \leqslant 1,\ \forall\, d = 1, \ldots, D$, recalling that $\phi_{d(k)} \equiv 1$ whenever $d(k)$ is null. Now, to complete the proof, let us show that for any arbitrary objective vector $(C_d, d = 1, \ldots, D)$, the linear program

$$\text{minimize} \left\{ \sum_{d=1}^{D} C_d \phi_d\colon\ \phi \in \Lambda \right\} \tag{25}$$

has an optimal solution for which ϕ_d is binary valued.

We solve (25) sequentially as follows. Consider any decision node $k \in \arg\max\{|B_k|\}$, and notice that the variables ϕ_d for $d \in J_k$ appear only in the single corresponding constraint for k in (24). Hence, there exists an optimal solution in which $\phi_{\hat{d}} = \phi_{d(k)}$, where $\hat{d} \in \arg\min_{d \in J_k}\{C_d\}$, and $\phi_d = 0,\ \forall\, d \in J_k - \{\hat{d}\}$. This eliminates the variables ϕ_d for $d \in J_k$ from (25) along with the associated constraint for this k in (24), where, whenever $B_k \neq \varnothing$ (so that $d(k)$ exists), we also update the objective coefficient $C_{d(k)}$ for the upstream (toward the root node) variable $\phi_{d(k)}$ according to $C_{d(k)} \leftarrow C_{d(k)} + C_{\hat{d}}$. Repeating this step, we will finally solve separable problems for decision nodes $k \in K$ having $B_k = \varnothing$, for which the reduced linear program is given as follows for some transformed objective coefficients $(\hat{C}_d, d \in J_k)$:

$$\text{minimize} \left\{ \sum_{d \in J_k} \hat{C}_d \phi_d\colon\ \sum_{d \in J_k} \phi_d = 1,\ \phi_d \geqslant 0,\ \forall\, d \in J_k \right\},$$

and for which there exists an optimal binary solution for ϕ_d, $d \in J_k$. By back-substituting these binary values for the recorded solutions for the downstream decision nodes, we will get a binary optimal solution for ϕ_d, $d \in J_k, \forall\, k \in K$. □

2.2. Further Properties of LP(Ω)

The next set of results lays the groundwork for composing our proposed global optimization strategy for solving Problem DTO. Henceforth, for any Problem P, we shall denote its optimal value as $v[P]$. Also, let $\bar{x} = (\bar{p}, \bar{l}, \bar{q}, \bar{r}, \bar{\phi}, \bar{w}, \bar{\psi}, \bar{\theta}, \bar{y}_1, \bar{y}_2, \bar{z}, \bar{\xi})$ represent an optimal solution to LP(Ω).

Proposition 5. $v[\text{LP}(\Omega)]$ *gives a lower bound for* $v[\text{DTO}(\Omega)]$. *Moreover, if* $\bar{x}$ *solves LP*(Ω) *and satisfies* (12k)–(12n) *and* (12r)*, then* $\bar{x}$ *also solves DTO*(Ω) *with the same objective value.*

Proof. The proof follows from the construction of LP(Ω), noting that under the hypothesis of the proposition and the validity of LMr, $r = 1, 2$, we also then have that (12c) holds true. □

Proposition 6. *Let* $\bar{x}$ *solve LP*(Ω). *Let* $(\hat{l}, \hat{q}, \hat{r}) \equiv (\bar{l}, \bar{q}, \bar{r})$ *and let* $\hat{p}$ *be computed by* (5e)*, i.e.,* $\hat{p}_i = \bar{g}_i/(1 + \bar{g}_i)$ *where* $\bar{g}_i = \exp\{a_{i0} - \sum_{m=1}^{M} a_{im}\bar{q}_{im}\},\ \forall\, i \in I_e$. *Furthermore, set* $\hat{\theta}_j \equiv \hat{l}_j \prod_{i \in S_{1j}} \hat{p}_i \prod_{i \in S_{2j}} (1 - \hat{p}_i), \forall\, j \in \tau$, *and let* $\hat{\phi}$ *be a binary optimal solution to the linear program composed in Proposition* 4 *as given by*

$$\widehat{LP}\colon\quad \textit{minimize} \left\{ \sum_{d=1}^{D} c_d \phi_d + \sum_{j \in \tau} \hat{\theta}_j \phi_{d(j)}\colon\ \phi \in \Lambda \right\}. \tag{26}$$

Then, $(\hat{p}, \hat{l}, \hat{q}, \hat{r}, \hat{\phi})$ *is a feasible solution to Problem DTO with objective value* $v[\widehat{LP}]$.

Proof. From (12q) (as in (9b)), we have $\ln(p_i^l) \leqslant \bar{y}_{1i} \leqslant \ln(p_i^u)$ and $\ln(1 - p_i^u) \leqslant \bar{y}_{2i} \leqslant \ln(1 - p_i^l)$. Thus, from (12g), $\ln(p_i^l) - \ln(1 - p_i^l) \leqslant \bar{y}_{1i} - \bar{y}_{2i} = a_{i0} - \sum_{m=1}^{M} a_{im}\bar{q}_{im} \leqslant \ln(p_i^u) - \ln(1 - p_i^u)$. Hence, since $\hat{p}$ has been computed by using the constraint (5e), we have

$$\ln\left(\frac{p_i^l}{1 - p_i^l}\right) \leqslant \ln\left(\frac{\hat{p}_i}{1 - \hat{p}_i}\right) = a_{i0} - \sum_{m=1}^{M} a_{im}\bar{q}_{im} \leqslant \ln\left(\frac{p_i^u}{1 - p_i^u}\right),$$

which results in $p_i^l \leqslant \hat{p}_i \leqslant p_i^u$. Thus, from the constraints of LP(Ω), we have that $(\hat{p}, \hat{l}, \hat{q}, \hat{r})$ is feasible to

(5b)–(5f), (5h), and (5i). Furthermore, by the structure of LM2 and Propositions 2 and 4, $\widehat{LP}$ defined by (26) then yields a binary optimal solution $\hat{\phi}$ that represents the best optimal completion $(\hat{p}, \hat{l}, \hat{q}, \hat{r}, \hat{\phi})$ to the foregoing partial solution. □

Proposition 7. *Let $\bar{x}$ solve LP(Ω) with objective value $v[LP(\Omega)]$. If each of the variable values $\bar{p}_i$, $\bar{l}_j$, and $\bar{\theta}_j$ equals either its corresponding lower or upper bound and $\bar{\phi}$ is binary valued, then $\bar{x}$ solves DTO(Ω) with objective value $v[DTO(\Omega)] = v[LP(\Omega)]$.*

Proof. From the construction of LP(Ω), it is sufficient to show that $\bar{x}$ satisfies (12c) and (12k)–(12n). For the generic case (13), if γ equals either of its bounds, then constraints (14a) and (14b) imply that $f = \ln(\gamma)$. Likewise, if each of the variable values $\bar{p}_i$, $\bar{l}_j$, and $\bar{\theta}_i$ equals either of its corresponding bounds, then (12k)–(12n) are satisfied. Moreover, as established for LMr, $r = 1, 2$, in §2.1.2 and Proposition 2, if $\bar{\phi}$ is binary valued, then (12c) is satisfied. Thus, $\bar{x}$ is feasible to DTO(Ω). □

Proposition 8. *Let Ω be such that $p_i^l = p_i^u, \forall\, i \in I_e$, and $l_j^l = l_j^u$, $\forall\, j \in \tau$. If $\bar{x}$ solves LP(Ω) with $\bar{\phi}$ being binary valued, then $\bar{x}$ also solves DTO(Ω).*

Proof. If $p_i^l = p_i^u$ and $l_j^l = l_j^u$, we have that $\theta_j^l = \theta_j^u$ by (7b). The proof now follows from Proposition 7. □

2.3. Optimality-Induced Valid Inequalities

To further tighten the model representation by legitimately reducing the search region as well as improving the underlying continuous LP relaxation, we introduce in this section certain valid inequalities (denoted VIs) that are implied by optimality (rather than feasibility) considerations.

Proposition 9. *There exists an optimal solution to Problem DTO satisfying the following inequalities*:

$$q_{im} \leqslant s_m \phi_{d(i)}, \quad \forall\, i \in I_e,\ m = 1, \ldots, M, \tag{27a}$$

$$r_{jn} \leqslant t_n \phi_{d(j)}, \quad \forall\, j \in \tau,\ n = 1, \ldots, N. \tag{27b}$$

Proof. First of all, note that Problem DTO has an optimal solution because it is bounded and feasible (the solution $(q, r) = (0, 0)$, with $p_i = p_i^u \equiv p_i^{u(3)}$, $\forall\, i \in I_e$, $l_j = l_j^u \equiv l_j^{u(4)}$, $\forall\, j \in \tau$, and any ϕ satisfying (5g) and (5j) gives a feasible solution). Now, for any $i \in I_e$, if $\phi_{d(i)} = 0$, then event i is inconsequential to the problem, and so we need not allocate any event-related resources to reduce p_i below p_i^u in (5e). Hence, we can set $q_{im} = 0$, $\forall\, m = 1, \ldots, M$, i.e., (27a) is valid. On the other hand, if $\phi_{d(i)} = 1$, then (27a) is again valid because it is implied by (5b). Likewise, (27b) is satisfied at an optimal solution because for any $j \in \tau$, if $\phi_{d(j)} = 0$, then consequence $j \in \tau$ does not arise (or does not impact the objective function), and we can therefore set $r_{jn} = 0$, $\forall\, n = 1, \ldots, N$; whereas if $\phi_{d(j)} = 1$, then (27b) is implied by (5c). □

Henceforth, we shall assume that the inequalities (27a) and (27b) are incorporated within Problem DTO, and hence within DTO(Ω) and LP(Ω), $\forall\, \Omega$. In addition, we shall perform the following variable fixings in Problem DTO(Ω) and LP(Ω) as prompted by Proposition 10 below. This strategy will become relevant in the sequel when we revise the bounds on the p- and l-variables in a branch-and-bound framework. Naturally, if any such fixings render a particular node subproblem infeasible, then we can fathom this node.

Proposition 10. *Consider any (node subproblem) DTO(Ω) predicated on a set of imposed bounds Ω (and incorporating the VIs* (27a) *and* (27b)*). Then,*

$$p_i^u < p_i^{u(3)} \ \Rightarrow\ \phi_d = 1, \quad \forall\, d \in B_i,\ \textit{for each } i \in I_e,$$

$$l_j^u < l_j^{u(4)} \ \Rightarrow\ \phi_d = 1, \quad \forall\, d \in B_j,\ \textit{for each } j \in \tau.$$

Proof. Consider any $i \in I_e$, and suppose that the currently imposed upper bound p_i^u on p_i satisfies $p_i^u < p_i^{u(3)}$. Then, by (12g), (12k), and (12l), we have that $q_{im} > 0$ for some $m \in \{1, \ldots, M\}$, and so, (27a) and (12r) imply that $\phi_{d(i)} = 1$. However, because the constraints (12c), (12i), and (12r) imply (19), we hence have $\phi_d = 1, \forall\, d \in B_i$. Similarly, for any $j \in \tau$, if the currently imposed upper bound l_j^u on l_j satisfies $l_j^u < l_j^{u(4)}$, then (12h) implies that $r_{jn} > 0$ for some $n \in \{1, \ldots, N\}$, and then (27b) and (19) yield $\phi_d = 1$, $\forall\, d \in B_j$. □

2.4. Upper-Bounding Scheme: Procedure UB

In this subsection, we use Proposition 6 to develop a method for computing upper bounds on Problem DTO based on the solution of any lower-bounding relaxation LP(Ω). (If any optimization problem is detected to be infeasible in this process, we abort the procedure.)

Procedure UB:

Step 1: Solve LP(Ω) (including constraints (27)) and obtain an optimal solution $\bar{x}$. Apply Proposition 6 to obtain the solution $(\hat{p}, \hat{l}, \hat{q}, \hat{r}, \hat{\phi})$.

Step 2: Fix $\phi = \hat{\phi}$, resolve LP(Ω) to obtain an optimum $\bar{x}$, and apply Proposition 6 to derive a possibly revised solution $(\hat{p}, \hat{l}, \hat{q}, \hat{r}, \hat{\phi})$. Repeat this step until no further improvement is obtained in the objective function value for Problem DTO.

Step 3: Fix $\phi = \hat{\phi}$, and use a nonlinear programming (local search) solver to optimize DTO(Ω), starting with the solution obtained at Step 2 as an initial solution. (We used SNOPT Version 7, Gill et al. 2005, for this purpose.) Output the resulting solution value as an upper bound on Problem DTO, and update the incumbent solution x^* and its objective value v^*, if necessary.

2.5. Range Reduction

The imposed range for each ϕ-, p-, l-, and θ-variable can be tightened by sequentially solving a pair of linear programs that minimize and maximize each variable in turn over the feasible region of LP(Ω), while additionally restricting the original objective function to take on values lesser than or equal to the best-known upper bound v^* obtained by Procedure UB. Following this range reduction process, the polyhedral outer approximations for the logarithmic relationships are constructed based on the revised bounds on the variables. Next, LP(Ω) is (re-)solved and the lower and upper bounds on Problem DTO are updated as possible. Note that at each node of the branch-and-bound tree, we invoke range reduction only once.

3. Global Optimization Branch-and-Bound Algorithms

We now design three alternative approaches to solve Problem DTO via DTO(Ω) and its relaxation LP(Ω). In *Algorithm A*, we utilize a specialized branch-and-bound process based on partitioning the hyperrectangle Ω, where the bounds on the variables $(\psi, \theta, y_1, y_2, z, \xi)$ are accordingly computed by (7e), (7b), (9b), (9b), (10b), and (11b), respectively. For any node subproblem DTO(Ω) that is associated with a particular Ω, we construct the relaxation LP(Ω) and solve it to compute a lower bound. Let $\bar{x}$ solve LP(Ω). If the conditions of Proposition 5 hold, we will have also solved subproblem DTO(Ω). Otherwise, we apply Procedure UB to find a feasible solution to Problem DTO and update the incumbent solution and the associated upper bound if possible, perform range reductions to update Ω and (12q), and as necessary, we branch at this node by partitioning Ω as follows:

Branching Rule A. While selecting the branching variable, priority is given to the ϕ_d-variables. Thus, we first check if $\bar{\phi}$ is binary valued. If not, then we define the set $K' = \{k \in K\colon\ (\bar{\phi}_d, d \in J_k)$ is not binary valued$\}$ and find $\hat{k} \in \arg\min_{k\in K'}\{|B_k|, \min_{d\in J_k} |\bar{\phi}_d - 0.5|\}$. We then branch on $\phi_{\hat{d}}$, where $\hat{d} \in \arg\min_{d\in J_{\hat{k}}}\{|\bar{\phi}_d - 0.5|\}$, by using the dichotomy that $\phi_{\hat{d}} = 1 \vee \phi_{\hat{d}} = 0$. Note that constraints (12b) and (12i) together with Equations (18) and (19) are invoked to fix additional (ϕ, w)-variables as possible whenever we branch on a ϕ-variable. On the other hand, if $\bar{\phi}$ is binary valued, then we find a variable p_i or l_j having the largest bounding interval according to

$$\max\{(p_i^u - p_i^l), (l_j^u - l_j^l),\ \forall i, j\}, \tag{28a}$$

where ties are broken by favoring the variable that gives the largest discrepancy in

$$\max\big\{|\bar{y}_{1i} - \ln(\bar{p}_i)|, |\bar{y}_{2i} - \ln(1-\bar{p}_i)|, |\bar{\xi}_j - \ln(\bar{l}_j)|,\ \forall i, j\big\}, \tag{28b}$$

where the first two terms in (28b) relate to p_i and the third term to l_j. If the identified term in (28) relates to p_i, then we branch on this variable by partitioning its interval according to the dichotomy that $p_i \in [p_i^l, (p_i^l + p_i^u)/2] \vee p_i \in [(p_i^l + p_i^u)/2, p_i^u]$. If the identified term corresponds to l_j, then we partition its interval according to the dichotomy that $l_j \in [l_j^l, (l_j^l + l_j^u)/2] \vee l_j \in [(l_j^l + l_j^u)/2, l_j^u]$.

Algorithm B is the same as Algorithm A, except that we also include θ_j in the partitioning process, as motivated by Proposition 7. In this case, given any imposed bounds $[\theta_j^l, \theta_j^u]$ on the variable θ_j, $\forall\, j \in \tau$, we update these bounds based on the implied bounds derived via (7b) according to

$$\left[\begin{array}{l} \theta_j^l \leftarrow \max\left\{\theta_j^l,\ l_j^l \prod\limits_{i\in S_{1j}} p_i^l \prod\limits_{i\in S_{2j}} (1-p_i^u)\right\} \\ \theta_j^u \leftarrow \min\left\{\theta_j^u,\ l_j^u \prod\limits_{i\in S_{1j}} p_i^u \prod\limits_{i\in S_{2j}} (1-p_i^l)\right\} \end{array}\right], \quad \forall\, j \in \tau.$$

Now, define Ω' as

$$\Omega' \equiv \big\{(p, l, \theta)\colon\ p_i^l \leqslant p_i \leqslant p_i^u,\ \forall\, i,\ l_j^l \leqslant l_j \leqslant l_j^u,\ \forall\, j,\ \theta_j^l \leqslant \theta_j \leqslant \theta_j^u,\ \forall\, j\big\}, \tag{29}$$

and let DTO(Ω') be identical to DTO(Ω), except that we primarily impose the bounds $(p, l, \theta) \in \Omega'$ and then compute the bounds on the variables (ψ, y_1, y_2, z, ξ) using (7e), (9b), (9b), (10b), and (11b), respectively. Whenever range reduction is performed, all these bounds are updated accordingly.

At each node of the branch-and-bound tree for Algorithm B, we proceed exactly as in Algorithm A except that we now solve the relaxation problem LP(Ω') for computing lower bounds, and also, for the partitioning scheme, the Branching Rule A is substituted by the following, where $\bar{x}$ solves LP(Ω'):

Branching Rule B. We apply the same scheme as in Branching Rule A in case $\bar{\phi}$ is not binary valued. Else, we find a variable that yields the maximum discrepancy according to

$$\max\big\{|\bar{y}_{1i} - \ln(\bar{p}_i)|, |\bar{y}_{2i} - \ln(1-\bar{p}_i)|, |\bar{\xi}_j - \ln(\bar{l}_j)|, |\bar{z}_j - \ln(\bar{\theta}_j)|, \forall\, i, j\big\}, \tag{30}$$

where ties are broken by favoring the variable having the largest bounding interval. If the maximum is attained by one of the first two terms, then we branch on the corresponding variable p_i by partitioning its interval according to $[p_i^l, \bar{p}_i] \vee [\bar{p}_i, p_i^u]$. Similarly, if the maximum is attained by the third or fourth term, we branch with respect to l_j or θ_j, partitioning their intervals as $[l_j^l, \bar{l}_j] \vee [\bar{l}_j, l_j^u]$ or $[\theta_j^l, \bar{\theta}_j] \vee [\bar{\theta}_j, \theta_j^u]$, respectively.

The third alternative, called *Algorithm C*, can be viewed as a combination of Algorithms A and B. Motivated by Propositions 7 and 8, we adopt the following partitioning scheme:

Branching Rule C. If $\bar{\phi}$ is not binary valued, then we apply Branching Rule A as before. Otherwise, we select a

variable having the largest bounding interval among the set of variables having at least a δ-deviation from their associated logarithmic functions, where $0 \leqslant \delta \leqslant 10^{-1}$. The value of δ changes through the optimization process depending on the number of variables that violate the exact corresponding logarithmic relations. Specifically, we select a variable p_i or l_j having the largest bounding interval according to

$$\max\{(p_i^u - p_i^l), (l_j^u - l_j^l),\ \forall i \in I',\ \forall j \in \tau'\}, \tag{31}$$

where the index sets I' and τ' are defined by

$$I' \equiv \{i\colon |\bar{y}_{1i} - \ln(\bar{p}_i)| > \delta\} \cup \{i\colon |\bar{y}_{2i} - \ln(1-\bar{p}_i)| > \delta\} \quad \text{and}$$

$$\tau' \equiv \{j\colon |\bar{\xi}_j - \ln(\bar{l}_j)| > \delta\}.$$

If both I' and τ' are empty, we set $\delta \leftarrow \delta/10^v$ for the smallest integer $v \geqslant 1$ such that at least one of these index sets becomes nonempty, and then apply (31). For the selected branching variable, we split its interval at the geometric mean, i.e., if the identified term in (31) corresponds to some p_i-variable, then we partition its interval according to $[p_i^l, \sqrt{p_i^l p_i^u}] \vee [\sqrt{p_i^l p_i^u}, p_i^u]$, and if it corresponds to some l_j-variable, we partition its interval according to $x[l_j^l, \sqrt{l_j^l l_j^u}] \vee [\sqrt{l_j^l l_j^u}, l_j^u]$.

Now we formally describe the branch-and-bound procedure Algorithm A for solving Problem DTO. At each stage s of this algorithm, $s = 0, 1, \ldots$, we define A_s as the set of nonfathomed, or *active nodes*. Each active node $a \in A_s$ is associated with a hyperrectangle Ω^a. A lower bound LB_a on an active node is obtained by solving the linear programming relaxation $\mathrm{LP}(\Omega^a)$ to yield $LB_a = v[\mathrm{LP}(\Omega^a)]$. At each stage s, we define the global lower bound on Problem DTO by

$$LB(s) \equiv \min\{LB_a\colon a \in A_s\}.$$

Whenever $\mathrm{LP}(\Omega^a)$ is solved for a node $a \in A_s$, we apply Procedure UB to possibly find a feasible solution for updating the upper bound for Problem DTO and perform range reductions as necessary. These bounds are used in concert with a least lower-bound node-selection rule and the aforementioned partitioning schemes as detailed below.

3.1. Branch-and-Bound Algorithm A for Problem DTO

Step 0: **Initialization.** Set $s = 0$, $A_s = \{0\}$, $a(s) = 0, a = 0$, and $\Omega^0 = \Omega$. Solve $\mathrm{LP}(\Omega^0)$ and let x^0 be the optimal solution obtained. Set $LB_0 = v[\mathrm{LP}(\Omega^0)]$. Apply Procedure UB of §2.4 with $\bar{x} \equiv x^0$ to derive an incumbent solution x^* with objective value v^*. Invoke the range reduction scheme of §2.5 to revise LB_0 and (x^*, v^*) if possible. If $LB_0 \geqslant v^*(1-\epsilon)$ for some optimality tolerance $\epsilon \geqslant 0$, then stop with the incumbent solution as (ϵ)-optimal to Problem DTO. Otherwise, proceed to Step 1.

Step 1: **Partitioning Step.** Invoke the Branching Rule A to partition the selected node $a(s)$ into two subnodes $a+1$ and $a+2$ with associated hyperrectangles Ω^{a+1} and Ω^{a+2}, respectively. Replace $A_s \leftarrow A_s \cup \{a+1, a+2\} - \{a(s)\}$.

Step 2: **Bounding Step.** Solve the relaxed problems $\mathrm{LP}(\Omega^{a+1})$ and $\mathrm{LP}(\Omega^{a+2})$ after fixing ϕ-variables if possible using Proposition 10. Apply Procedure UB to the solutions found at each node and update the incumbent solution if possible, and perform range reductions as necessary.

Step 3: **Fathoming Step.** Fathom any nonimproving node and update $A_{s+1} \leftarrow A_s - \{a \in A_s\colon LB_a \geqslant v^*(1-\epsilon)\}$. Increment s by 1.

Step 4: **Termination Check and Node-Selection Step.** If $A_s = \varnothing$, stop with the incumbent solution as an $\epsilon-$optimum. Otherwise, select a node $a(s) \in \operatorname{argmin}\{LB_a\colon a \in A_s\}$ and return to Step 1.

Proposition 11 (Main Convergence Result). *Algorithm A (with $\epsilon = 0$) either terminates finitely with the incumbent solution as an optimum to DTO, or else an infinite sequence of stages is generated such that along any infinite branch of the tree, any accumulation point of the (p, l, q, r, ϕ)-variable part of the sequence of linear programming relaxation solutions solves Problem DTO.*

Proof. The case of finite termination is clear. Hence, suppose that an infinite sequence of stages is generated. Consider any infinite branch of the tree having a nested hyperrectangle sequence $\Omega^{a(s)}$ for s belonging to some index set S. For each stage s, we have $LB(s) = LB_{a(s)} = v[\mathrm{LP}(\Omega^{a(s)})]$, $\forall s \in S$. For each node $a(s)$, let $x^{a(s)}$ solve $\mathrm{LP}(\Omega^{a(s)})$. By the compactness of the feasible region, there exists a convergent subsequence for $\{x^{a(s)}, \Omega^{a(s)}\}_{s\in S}$. Without loss of generality, assume that $\{x^{a(s)}, \Omega^{a(s)}\}_{s\in S} \to (x^*, \Omega^*)$. We must show that the $(p^*, l^*, q^*, r^*, \phi^*)$-variable part of x^* solves Problem DTO.

Note that $LB_{a(s)} = \min\{LB_a\colon a \in A_s\} \leqslant v[\mathrm{DTO}]$, $\forall s \in S$, which is also preserved in the limit:

$$v^* \equiv \lim_{s\to\infty,\, s\in S} LB_{a(s)} \leqslant v[\mathrm{DTO}]. \tag{32}$$

Because we can only branch on the ϕ-variables finitely often, we have that $\phi^{a(s)}$ is binary valued for all $s \in S$ sufficiently large. Hence, along the infinite branch, at least one of the p_i- or l_j-variables is chosen as the branching variable infinitely often. Because we bisect the corresponding interval at each step, the length of the interval for such a variable converges to 0. Therefore, according to Branching Rule A, we have in the limit as $s \to \infty, s \in S$ that $p_i^{*l} = p_i^{*u}$, $\forall i$, and $l_j^{*l} = l_j^{*u}$, $\forall j$. By Proposition 8, we thus have that x^* is feasible to DTO with objective function value $v^* \equiv \lim_{s\to\infty, s\in S} LB_{a(s)}$, and so $v^* \geqslant v[\mathrm{DTO}]$. Together with (32), we have that x^* solves DTO with objective function value v^*. □

3.2. Branch-and-Bound Algorithm B for Problem DTO

This alternative branch-and-bound procedure is the same as Algorithm A, with the exception of the branching variable selection at each stage s along with the partitioning

of the hyperrectangle Ω' associated with each active node $a \in A_s$, where Ω' is defined by (29). The following result establishes the global convergence of Algorithm B.

PROPOSITION 12. *Similar to the main convergence result of Algorithm A stated in Proposition* 11, *Algorithm B* (*with* $\epsilon = 0$) *solves Problem DTO.*

PROOF. The case of convergence in a finite number of steps is clear. For the case of infinite stages, taking any infinite branch of the tree where the stages s are indexed by a set S, we have as in the proof of Proposition 11 that $\{x^{a(s)}, \Omega'^{a(s)}\}_{s\in S} \to \{x^*, \Omega'^*\}$, and that $LB(s) = LB_{a(s)} = v[\mathrm{LP}(\Omega'^{a(s)})] \leqslant v[\mathrm{DTO}], \forall\, s \in S$, which also holds true in the limit:

$$v^* = \lim_{s\to\infty, s\in S} LB_{a(s)} \leqslant v[\mathrm{DTO}]. \tag{33}$$

We now show that x^* is feasible to Problem DTO. Along the convergent subsequence, for s large enough, $\phi^{a(s)} = \phi^*$ is binary valued, and so some p_i-, l_j-, or θ_j-variable is partitioned infinitely often at stages $s \in S_1$, say, where $S_1 \subseteq S$. From Branching Rule B, this variable equals one of its bounds in the limit, and therefore, the discrepancy related to it in (30) approaches zero. Because this variable has the maximum discrepancy in (30), $\forall\, s \in S_1$, the discrepancies related to the other p-, l-, and θ-variables also approach zero as $s \to \infty, s \in S_1$. Consequently, by Proposition 7, x^* satisfies constraints (12c) and (12k)–(12n) with $\phi^*_d \in \{0, 1\},\ \forall\, d \in D$. Hence, x^* is feasible to DTO with objective value v^*, so that $v^* \geqslant v[\mathrm{DTO}]$. Together with (33), we have that x^* solves DTO with objective value v^*. □

3.3. Branch-and-Bound Algorithm C for Problem DTO

As described earlier in the section, this procedure follows the same scheme as that of Algorithms A and B, while adopting a combination of these two methods for selecting a branching variable and partitioning its interval, and including the alternative option of splitting the interval of the selected variable at its geometric mean. As such, its global convergence proof follows identically to that in Propositions 11 and 12 above and is omitted for the sake of brevity.

4. Computational Results

In this section, we study the effectiveness of the proposed branch-and-bound procedures for solving Problem DTO. We begin by considering the gas-line rupture application illustrated in Figure 1 and solve it using the proposed algorithms as well as using the commercial global optimization software BARON, Version 8.1.5 (Tawarmalani and Sahinidis 2005). We further analyze the sensitivity of the solution with respect to the budget and resource availability restrictions, and also with respect to variations in the logit and loss function parameters.

Next, we explore the best combination of algorithmic features including the various branching variable selection schemes and partitioning strategies. We also investigate the performance of the different proposed linearization methods and techniques for generating tangential supports, and assess the effect of incorporating range reductions and VIs on CPU time. In addition, we examine the direct implementation of the software BARON to solve the original problem formulation DTO (given by (5)) as well as its transformed version DTO(Ω) (given by (12)), in comparison with our best proposed procedure. Runs with BARON have been made on a remote 1.6 GHz Intel Pentium M processor running Linux (courtesy of N. V. Sahinidis), whereas the proposed procedures have been implemented on a local 2.33 Ghz Intel Pentium Xeon processor running Windows. (All runs with BARON in this paper have been made with default settings with increased memory workspace and by pregenerating the GAMS problem file prior to inputting this into the BARON solver.)

4.1. Gas-Line Rupture: Illustrative Example

Consider the scenario in which, in the aftermath of a gas leak (represented by node 0), cascading sequences of probabilistic events and decisions result in consequences as depicted in Figure 1. For this specific gas-line rupture application, we assume that five event-related and five consequence-related resources are available to prevent failures and to alleviate consequences. The model coefficients a_{im} and b_{jn} were generated uniformly in the respective intervals $[0.05, 0.1]$ and $[100, 500]$. Additionally, we set $a_{i0} = \ln(0.05/(1 - 0.05)), \forall\, i$, and generated $b_{j0}, \forall\, j$, randomly on $[10{,}000 + l^l_j, 10{,}000 + 2l^l_j]$. The total available event-related and consequence-related resources were set to $s_m = 10, \forall\, m$ and $t_n = 20, \forall\, n$, respectively, where the corresponding per-unit costs of allocating resources, c_{im} and d_{jn}, were generated uniformly on $[20, 40]$. The total available budget for resource allocation was initially taken as $\beta = 3{,}500$. The lower bound on $l_j, \forall\, j \in \tau$ was determined depending on the scenario produced by the unique path connecting the root node to the leaf node j. More specifically, a contribution to l^l_j of $5{,}000(2.5)^k$ was set to be incurred for each failure event on the path from node 0 to the leaf node, where k is the nodal distance between the failed event and the leaf node. The lower bound on l_j for any leaf node that has no failure event along the path to it from node 0 was set to 500. The upper bound was determined as $l^{u(4)}_j, \forall\, j \in \tau$, whereas the lower and upper bounds on p_i were set to 0.0001 and $p^{u(3)}_i$, respectively, $\forall\, i \in I_e$. Moreover, the direct cost for each selected decision alternative $d = 1, \ldots, D = 9$ was generated randomly on $[100, 200]$.

Using the above data, we first explored the effect of uncertainty in logit and loss function parameters on the failure probabilities and loss amounts, and hence on the total expected loss in the system. Toward this end, suppose that a policy maker estimates function parameters

$a_{im} = \bar{a}_{im}, \forall i, m$, and $b_{jn} = \bar{b}_{jn}, \forall j, n$, with incomplete information and, accordingly, determines optimal decision alternatives, ϕ^*_{est}, and resource allocations, q^*_{est} and r^*_{est}. Now, assume that there exists up to a $\pm 5\%$ error, say, in estimating the logit and loss function parameters, so that their actual values (uniformly generated) are given by $\bar{a}_{im} \pm 0.05\bar{a}_{im}$ and $\bar{b}_{jn} \pm 0.05\bar{b}_{jn}$, respectively. Consequently, by substituting the implemented policy decisions $(\phi^*_{est}, q^*_{est}, r^*_{est})$ within the model functions represented by the actual parameter values, we can compute the corresponding failure probabilities and loss amounts, yielding an actual realized total expected loss of z_{act}. On the other hand, if these actual parameters had been known a priori, we could have obtained the optimal expected loss, z^*_{act}, by applying the proposed algorithm. The resulting percentage optimality gap, $100(z_{act} - z^*_{act})/z^*_{act}$, provides an indication of the sensitivity of the quality of the derived solution to the function parameters and also reflects on the value of perfect information. We utilized parameter estimation errors of $1\%, \ldots, 5\%$, and replicated each case 10 times. Figure 2 displays the average percentage optimality gaps obtained for the different uncertainty levels, where the expected loss is in the order of 1,775. For error levels of $1\%, \ldots, 5\%$ in estimates, the average percentage optimality gaps are $1.3\%, 1.6\%, 2.7\%, 3.7\%$, and 6.0%, respectively. Note that the lower and upper bounds on failure probabilities and the losses at leaf nodes, as well as the magnitudes of the logit and loss function parameters influence this computed percentage optimality gap, and the results shown here are based on reasonable parameter ranges and are meant to serve only as an illustration of the types of studies a decision maker might undertake to assess the sensitivity of the quality of solutions being considered for implementation with respect to possible errors in parameter estimations.

Next, using five levels of the available budget, $\beta \in \{2{,}000, \ldots, 6{,}000\}$, and four levels of event-related resources, $s \in \{20, \ldots, 50\}$, we computed the optimal expected loss (objective function value) and the variance of loss (as given by Equation (6)). The results are displayed in Figure 3. As a point of interest, the minimum expected loss and the associated variance for the case of unlimited resources are depicted by horizontal lines in the corresponding graphs. As expected, the objective function value and the variance of loss improve significantly with initial increments in budgetary and event-related resources, but further resource increases result in diminishing marginal returns. Note that the exceptional variance increase at $(\beta, s) = (6{,}000, 40)$ is due to the discrete restructuring of the tree by virtue of ϕ_2 being activated, where, prior to this, the decisions ϕ_3 and ϕ_6 were active. Hence, the decline in variance to its limiting value as resources increase need not be monotone due to the discrete nature of the problem.

As a point of interest, all of the foregoing problems were solved in about two CPU seconds using either the proposed algorithm or BARON.

Figure 2. Effect of estimation errors in logit and loss function parameters on the expected loss.

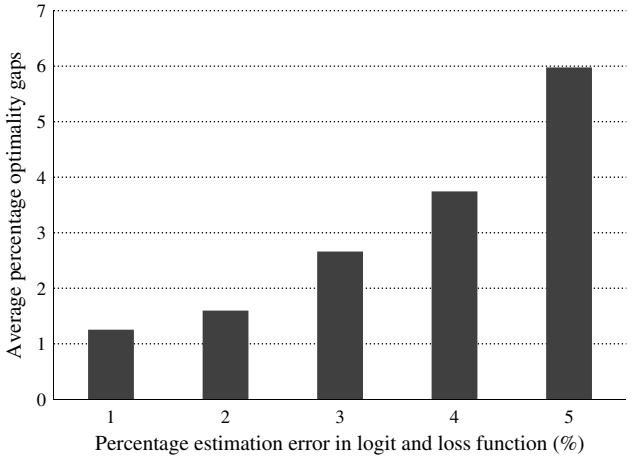
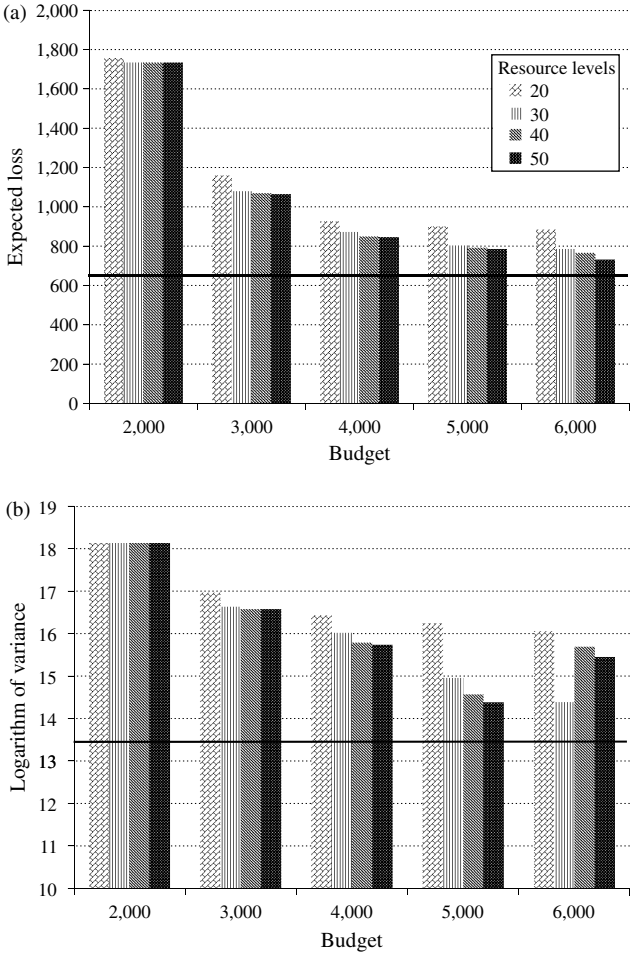


Figure 3. Effect of budget and event-related resources on expected loss and variance.



4.2. Random Test Cases

We next generated random decision tree problem instances to assess the effectiveness of the proposed solution

techniques as well as that of the global optimization software BARON (Version 8.1.5). The two inputs provided to the random tree generation process are the desired *decision node density*, as defined by the ratio of the number of decision nodes to the total number of nodes, and the size of the tree as measured by its maximal depth. To complete the instance specification, we randomly generated the logit coefficients a_{im} and b_{jn}, the cost coefficients c_{im} and d_{jn}, and the decision costs c_d uniformly over the respective intervals $[0.5, 1]$ and $[1{,}000, 5{,}000]$, $[20, 40]$ and $[20, 40]$, and $[2, 4]$. The lower bounds on l_j were determined as in §4.1, except that the values 5,000 and 500 therein were replaced by 1,000 and 10, respectively. We set b_{j0} equal to the sum of the lower bound on l_j and a uniformly generated random variable over the interval $[100{,}000, 150{,}000]$ for the first 10 instances. For the next 10 instances, b_{j0} were generated over the interval $[100{,}000, 200{,}000]$. The budget and the decision node density were, respectively, selected as 3,000 and $0.18 - 0.38$ (specified desired value $= 0.3$) for the instances numbered 1–5 and 11–15, and as 1,500 and $0.56 - 0.74$ (specified desired value $= 0.7$) for the instances numbered 6–10 and 16–20. The first 10 instances have a depth of five and are preliminarily used for evaluating the performance of eight algorithmic strategy combinations, whereas the depth of the remaining instances is six. The resulting number of leaf nodes range from 22–42 and 36–106 for the test instances having a maximum depth of five and six, respectively.

In the following runs, we define the *base-case approach* as the one that incorporates the linearization scheme LM2 (§2.1.2), the valid inequalities VIs (§2.3), and invokes range reduction (§2.5), while generating four tangential supports for each functional form $f = \ln(\gamma)$ using the BES ($H = 4$) strategy, and utilizes an optimality gap tolerance of $\epsilon = 0.001$.

Using the base-case approach, we first studied the efficiency of the three branching variable selection rules (A, B, and C) together with the three splitting rules (arithmetic mean, geometric mean, and the current optimal value), for a total of eight combinations as displayed in Table 1, where the average CPU time values across rows are displayed in bold in the final column. (Note that Rule A, combined with the current optimal value partitioning strategy, is not included because the current optimal value could be at the lower or upper bound of the selected branching variable, which would consequently not produce valid child-nodes.) Observe also that Algorithm A is defined by the branching variable selection rule A and the arithmetic mean splitting rule (Combination 1), whereas Algorithm B is defined by the branching variable selection rule B and the current optimal value splitting rule (Combination 5). Algorithm C is (principally) defined by the branching variable selection rule C and the geometric mean splitting rule (Combination 7).

The branching variable selection rule C and the arithmetic mean splitting rule outperformed the other combinations, and therefore this composite strategy was used for further evaluating various algorithmic variants as discussed next. (Algorithm C, defined by Combination 7, comes a close second, and is also further evaluated in subsequent runs. In fact, with BES($H = 5$) used in lieu of BES($H = 4$), the mean CPU times for Combinations 6 and 7 were 26.4 and 26.0 CPU seconds, respectively.)

We next performed a series of experiments in which one of the features in the base case was varied to ascertain the effects of the different implementation strategies. The results indicated that utilizing the linearization method LM2, generating $H = 4$ tangential supports via BES($H = 4$), and invoking range reduction and valid inequalities outperformed other alternatives. (The use of VIs reduced the CPU time by 13% over the same strategies without VIs; detailed tables are omitted for the sake of brevity.) Note that the relative efficiency of the linearization methods depends on the decision node density. Whereas both linearization schemes were comparable for the first five instances having a low decision node density (LM2 consumed 2.5% lesser effort), we observed a 13% improvement in the CPU time for the next five instances having a high decision node density when using LM2 versus LM1. This is to be expected because higher decision node density instances involve more complex binary substructures in the model, which therefore benefit more substantially by using the improved representation LM2 over LM1. We use these settings in further runs with our proposed algorithm, hereinafter referred to as *Algorithm D* (essentially, Combination 6 in Table 1). We also explore the efficiency

Table 1. Performance (CPU times) of eight algorithmic strategy combinations.

	Branching variable		Problem instance										
	Selection	Splitting	1	2	3	4	5	6	7	8	9	10	Avg.
1 (Alg. A)	Rule A	Arith.	10.8	150	8.6	31.4	5.2	16.3	11.9	8.8	22.9	15.3	**28.1**
2	Rule A	Geom.	10.4	139.8	8.2	27.4	5.1	16.4	10.7	9	22.6	14.8	**26.4**
3	Rule B	Arith.	11.6	132.8	9.8	71.3	11.3	16.6	11.8	8.8	22.9	14.8	**31.2**
4	Rule B	Geom.	10.9	187.7	12.1	142.8	9.8	16.3	11.4	8.9	22.8	15	**43.8**
5 (Alg. B)	Rule B	Current	14.4	215.2	12.3	124.2	10.6	15.9	11.7	9.1	22.6	14.9	**45.1**
6 (Alg. D)	Rule C	Arith.	10.1	115.1	8.4	26.4	5.1	16.3	11.3	8.8	23	14.8	**23.9**
7 (Alg. C)	Rule C	Geom.	10.4	125	8	25.7	5	16.3	10.7	8.9	22.8	14.6	**24.7**
8	Rule C	Current	10.5	157.5	8.2	27.5	4.9	16.6	11.4	8.8	22.6	15	**28.3**

Table 2. Results for BARON using the original and the transformed DTO formulations, vs. Algorithm D.

	BARON: Original DTO (5)		BARON: Transformed DTO (12) with LM2		BARON: Transformed DTO (12) with LM2 and VIs (27)		Proposed Algorithm D	
Problem instance	CPU time	Optimality gap (%)	CPU time	Optimality gap (%)	CPU time	Optimality gap (%)	CPU time	Optimality gap (%)
1	15.9	2.4*	1,800	50.2	99.5	0	9.6	0
2	1,800	30.2	1,800	59.7	1,800	61.1	108.1	0
3	2.6	1.8*	1,800	29.4	481.7	0	8.3	0
4	25.8	0	1,800	59.1	512.2	0	26.0	0
5	4.3	0	1,800	19.7	56.2	0	4.8	0
Avg.	**369.7**		**1,800**		**589.9**		**31.4**	
6	29.5	0	1,800	29.5	238.7	0	15.6	0
7	4.3	0	1,800	36.3	166.6	0	10.8	0
8	293.0	0	1,800	27.9	125.7	0	9.7	0
9	130.9	0	1,800	50.0	1,800	48.6	21.4	0
10	71	0	1,800	38.5	172.4	0	15.2	0
Avg.	**105.8**		**1,800**		**500.7**		**14.5**	
11	28	0	1,800	50	483.4	0	22.1	0
12	17.4	0	1,366	0	429.7	0	16.2	0
13	395.4	0	1,800	46.8	1,800	22.8	30.1	0
14	24	0	1,800	62.3	1,800	24.1	109.7	0
15	1,800	29.4	1,800	62.2	1,800	n/a	23.4	0
Avg.	**453**		**1,713.2**		**1,262.6**		**40.3**	
16	541.6	0	1,800	26.6	1,800	17.5	81	0
17	406.1	1.4*	1,800	44.9	1,454.9	0	37.3	0
18	78.5	0	1,381	0	474.9	0	20.9	0
19	180.2	0	1,800	100	340	0	17.2	0
20	12.1	0	403.6	0	58.4	0	6.7	0
Avg.	**243.7**		**1,436.9**		**825.7**		**32.6**	

*Terminated with a suboptimal solution having the specified optimality gap with respect to the optimum found by Algorithm D.

of Combination 7 (*Algorithm C*) along with Algorithm D because the difference in performance between these two procedures is not significant, as displayed in Table 1.

The decision tree optimization problem can alternatively be directly solved by applying off-the-shelf commercial software to either the original model (5) or to the better-structured equivalently transformed formulation (12). The global optimization software BARON was therefore utilized to solve problem instances modeled alternatively as in (5) or (12) using LM2, as well as to solve problems modeled via (12) along with both LM2 and the proposed VIs (27), to assess the direct solvability of the enhanced models using standard software. Table 2 displays the results obtained in comparison with Algorithm D, where the average CPU times for each set of five test instances are highlighted in bold. The optimality gap for each case is computed as $(UB - LB)/UB$, where UB and LB are the respective upper and lower bounds obtained on the optimal objective function value at termination. Using the original DTO formulation (5), BARON terminated prematurely for two of the 20 test problems (with an average optimality gap of 29.8%), whereas the proposed algorithm optimized all the problems within the specified CPU time limit of 1,800 seconds. Furthermore, BARON terminated with suboptimal solutions within the time limit for Instances 1, 3, and 17, with respective optimality gaps of 2.4%, 1.8%, and 1.4%. On average, it took 293 CPU seconds for BARON to solve Problem DTO (including the early termination cases), whereas the same set of problems were optimized in 29.7 CPU seconds with the proposed algorithm (yielding an 89.9% savings in effort). Excluding the early termination cases, the average CPU times were 122.4 seconds for BARON versus 27.2 seconds for the proposed algorithm (yielding a 77.8% savings in effort). As a point of interest, Algorithm C solved these instances in 30.6 CPU seconds on average, where this value decreased to 27.6 CPU seconds when the premature termination cases are excluded. Moreover, BARON solved only 3 out of the 20 problem instances to optimality using the transformed DTO formulation (12) with LM2, which may be due to the increase in the number of variables as well as the number of nonlinear constraints. However, when we further invoked the valid inequalities VIs within the formulation (12), BARON managed to solve 14 out of the 20 problem instances using this enhanced model representation formulation.

5. Summary and Conclusions

This paper addresses the strategic reduction of risk by allocating available resources to reduce failure probabilities of system safety components and subsequent losses related to final outcomes, as well as by selecting optimal strategic decisions or system design alternatives in

the context of a hazardous event. Using a novel decision tree approach, the problem was modeled as a nonconvex mixed-integer 0-1 factorable program. We developed a specialized branch-and-bound algorithm in concert with polyhedral outer approximations, valid inequalities, alternative linearization schemes, and range reduction strategies, and established its theoretical convergence to global optimality.

Among the various algorithmic variants tested, we advocate the strategy that implements branching variable selection rule C along with the arithmetic (or geometric) mean splitting rule, while utilizing the linearization method LM2 (6% CPU time reduction over LM1 on average, but 13% reduction on average for relatively higher decision node density problems); generating tangential supports via the BES($H = 4$) scheme; invoking the range reduction mechanism (all instances were solved to ϵ-optimality with range reduction compared to 100% early termination without range reduction), and incorporating the derived valid inequalities (13% CPU time reduction on average). In our computational experience, this proposed algorithmic approach outperformed the commercial software BARON (Version 8.1.5), yielding a more robust performance along with an 89.9% savings in effort on average.

This research can be extended in several directions. For instance, instead of minimizing the overall risk, we could minimize the maximum risk, or optimize the mean-variance based risk function for the system using Equations (5a) and (6). Additionally, as explained in Remark 2, we can use a diminishing marginal return loss function in lieu of the linear correspondent given by (4), as well as replace Bernoulli events with more general multistate events. Furthermore, as an algorithmic extension, we could explore a GUB-based partitioning strategy whenever we branch on a ϕ-variable. Finally, it is of interest to investigate specific applications of the proposed generic DTO problem framework in event-decision contexts that arise in various areas such as homeland security and health care.

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